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Palaeomagnetism and K–Ar ages of Lower Ordovician and Upper Silurian–Lower Devonian rocks from north-west Argentina

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Summary. In this paper we present palaeomagnetic data from 87 hand samples collected in a sequence of tuffs and shales (Suri Formation) of Llanvirnian age, exposed in north-western Argentina (27° 47′ S, 68° 06′ W). After cleaning, the majority of samples showed reversed polarity and yielded a palaeomagnetic pole at 5.9° E, 8.5° S ($\alpha_{95} = 5.9°$). They also showed reversals of declination and inclination at the top of the sequence, which we have associated with geomagnetic excursions. Whole rock K–Ar age determinations suggest an age older than 416 ± 25 Myr for the Suri rocks. The predominant reversed stable remanence of these rocks is consistent with the reversed polarity reported for Early Llanvirnian rocks from USSR. The palaeomagnetic pole for the Suri Formation is consistent with the interpretation that Gondwana was a single unit in Early Palaeozoic times.

Palaeomagnetic data from 27 hand samples collected from 10 igneous units of Late Silurian–Early Devonian age (Ñuñorco Formation), exposed in the same area, are also given. The majority of the igneous units showed reversed polarity after cleaning. The positions of VGP's for the Ñuñorco igneous units are scattered and they are not used for geodynamic interpretations. Whole rock K–Ar age determinations suggest ages of 416 ± 25 and 360 ± 10 Myr for two igneous units of the Ñuñorco Formation.

1 Introduction

The palaeomagnetic and radiometric studies of Early Palaeozoic rocks from Argentina are included in Project No. 44 (Lower Palaeozoic of South America) of the International Geological Correlation Programme. The palaeomagnetic data obtained in these studies are also useful in defining the geodynamic evolution of South America in Early Palaeozoic times. Vilas & Valencio (1978a) have reported the palaeomagnetic data and the isotopic

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age for a sequence of Late Ordovician rocks from western Argentina. In this paper we present palaeomagnetic data and K–Ar ages for a sequence of tuffs and shales (Suri Formation), assigned to the Early Ordovician, and igneous rocks (Ñuñorco Formation), assigned to the Late Silurian to Early Devonian period, exposed in north-western Argentina. We also compare the palaeomagnetic poles for these formations with the positions of poles from the other Gondwanic continents.

2 Geological evidence and sampling

Tuffs and shales (Suri Formation) and igneous rocks (Ñuñorco Formation) are exposed close to Puesto Chaschuil in the western part of the Province of Catamarca, Argentina (27° 47′ S, 68° 06′ W, Fig. 1). The tuffs and shales, named in chronological order, included in the Suri Formation (Turner 1967), show incipient metamorphism and are characterized by the presence of fossil assemblages (trilobites), from which Harrington & Leanza (1957) have suggested a Llanvirnian age for the formation. Recently, Maisonave (1973) has included the tuffs in an independent formation which he named Las Planchadas. In this paper we follow the classification suggested by Turner for these rocks.

Since the marine sedimentation of the Suri Formation, at least three igneous episodes have occurred (Turner 1967); Turner suggested an age younger than Llanvirnian and older than Carboniferous for these igneous rocks. Maisonave (1973) included the igneous rocks formed during these episodes in the Ñuñorco Formation and in this paper we follow his classification. Maisonave suggested an age between Late Silurian and Early Devonian for the Ñuñorco rocks.

The tuffs and shales of the Suri Formation were sampled in a rock sequence; 55 hand samples were collected (23 from tuffs and 32 from shales, Fig. 1). Samples were collected at different stratigraphic levels in the sequence (Table 2, Fig. 4), covering a stratigraphic interval of about 13 m tuffs and 16 m shales. Observations of the bedding plane were carried out along the sequence.

The igneous rocks of the Ñuñorco Formation were sampled at eight different sites (units A, B, C, D, F, G_1, G_2, H and I), corresponding to different igneous episodes of the Ñuñorco Formation (granular and granophyric facies and dykes and sills associated with both facies). Samples were also collected from the contact baked (wall rock, unit E) by the intrusion of dyke D and from a basaltic dyke (unit J) in the tuffs of the Suri Formation. This last unit is assigned to La Ojota Formation, which is thought to be Middle Silurian, based upon the evidence of stratigraphic relations (Maisonave 1973). Thirty-three samples were collected from these igneous units (Table 3).

Hand samples were oreinted using both sun and magnetic compasses.

3 Potassium–argon determinations

Whole potassium–argon age determinations were carried out on samples collected from tuffs of the Suri Formation, igneous rocks of the Nuñorco Formation and the dyke of La Ojota Formation. The study was carried out in the Instituto de Geocronología y Geología Isotópica (Universidad de Buenos Aires). The techniques used were those of Linares & Valencio (1975). The K–Ar data are shown in Table 1.

The ages obtained for the igneous rocks from the Ñuñorco Formation are consistent with its geological data.

In view of the palaeontological data (trilobites of Llanvirnian age, Early Ordovician), the ages obtained from the tuffs of the Suri Formation are considered to be anomalously

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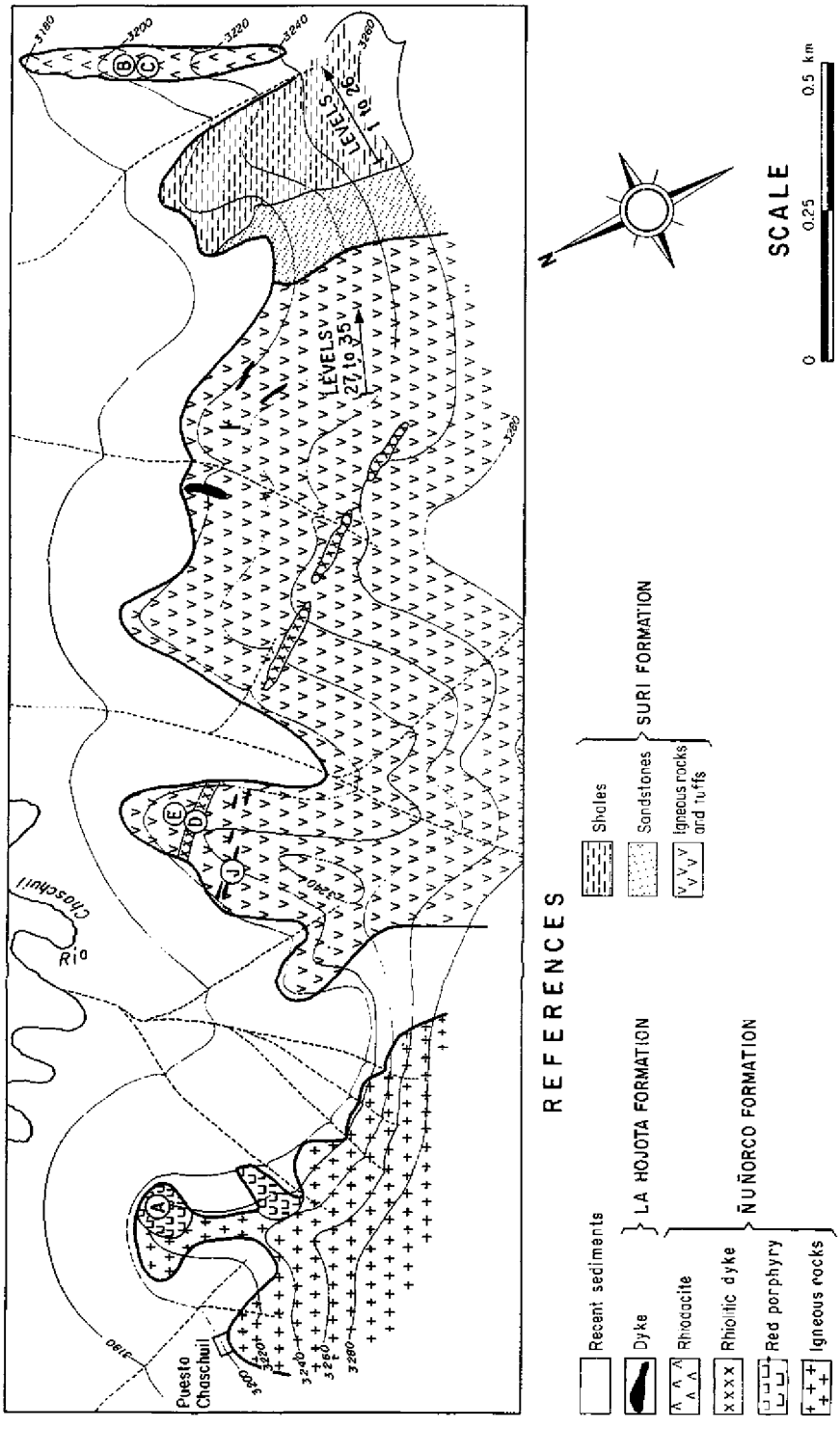


Figure 1. Schematic geological map close to Puesto Chaschuil (Patiño Douce 1975) showing sampling sites for the palaeomagnetic and radiometric studies.

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Table 1. K–Ar ages for the tuffs of the Suri Formation and igneous rocks of the Ñuñorco and La Ojota Formations exposed close to Puesto Chaschuil, Province of Catamarca, Argentina.

Formation	Rock	Unit	*K* (per cent)	Ar^{40}_{rad} 10^{-10} mol g^{-1}	Ar^{40}_{atm} (per cent)	Age (Myr)
La Ojota	Dyke	J	1.85	10.600	11.8	297 ± 10
Ñuñorco	Rhyolitic dyke	D	6.52	44.903	5.3	360 ± 10
Ñuñorco	Rhyodacite	B	0.045	0.373	67.0	416 ± 25
Suri	Tuff	29	1.69	9.707	13.5	304 ± 10
Suri	Tuff	27	0.86	5.773	13.0	350 ± 10

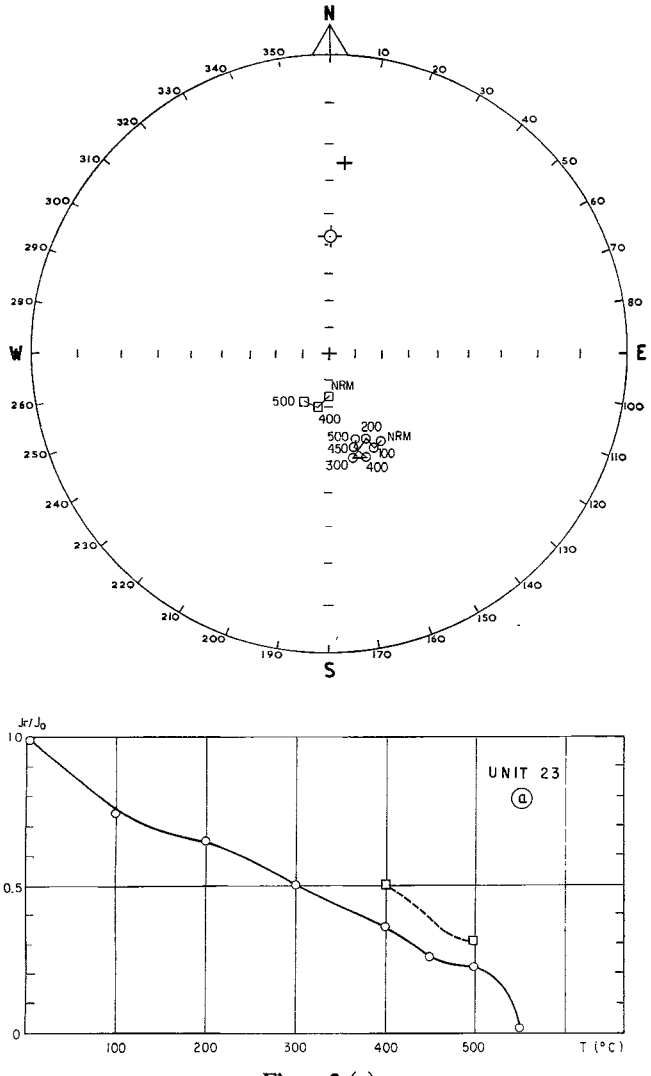


Figure 2 (a)

Figure 2. Changes in direction (referred to the palaeohorizontal) and intensity of NRM of the pilot specimen (○) and one other cylinder (□) from one sample from Suri shales in response to (a) thermal demagnetization and (b) Suri tuffs in response to ac demagnetization. Solid symbols indicate downward dipping directions. The directions of the present geomagnetic field and of the axial dipole are shown by + and ⊹ respectively.

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affected by argon loss. The radiometric age of the rhyodacitic unit (B) from the Ñuñorco Formation supports this view.

The dyke of La Ojota Formation yields an age of 297 ± 10 Myr, indicating that it is Late Carboniferous.

4 The palaeomagnetic study

One hundred and fifty-nine specimens (cylinders of 2.5 cm diameter and height) were cut. Two spinner magnetometers were used to measure the remanent magnetization of these specimens. Detailed alternating field demagnetization (ac) was used to isolate the stable remanence of one pilot specimen of each sample from the tuffs of the Suri Formation and the igneous rocks of the Ñuñorco and La Ojota Formations (Valencio, Vilas & Mendía 1977). Thermal demagnetization was used to isolate the stable remanence of one pilot

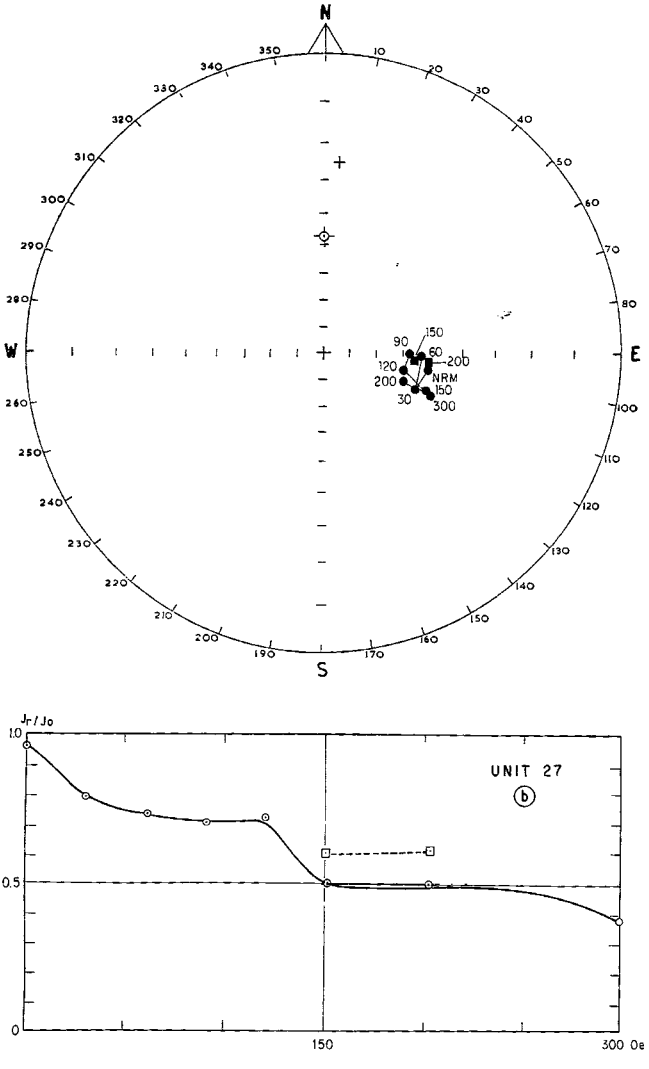


Figure 2 (b)

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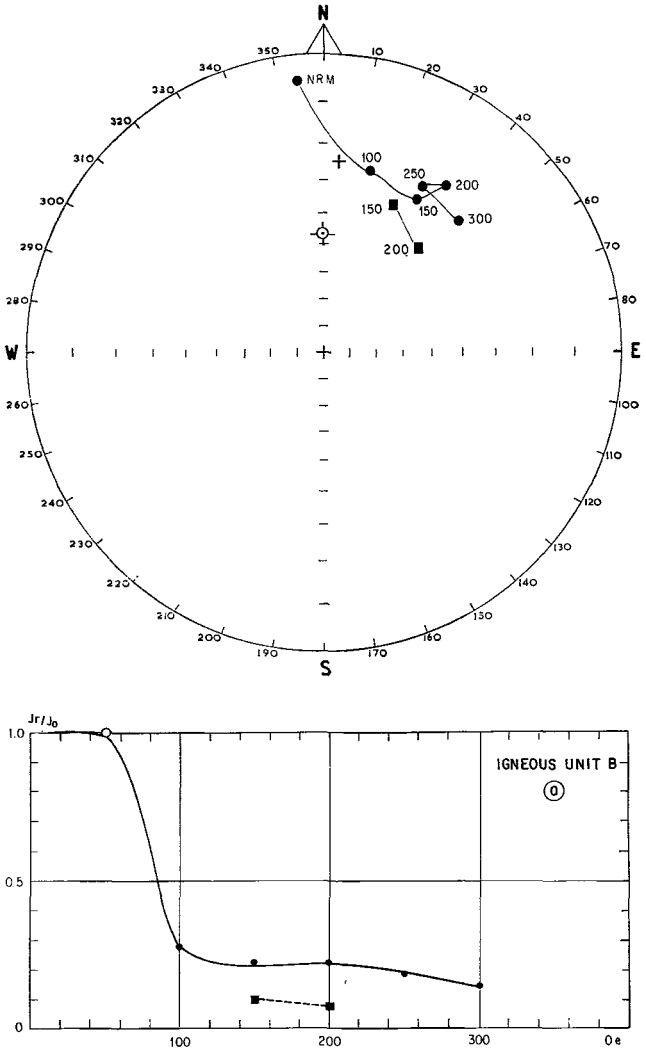


Figure 3 (a)

Figure 3. Changes in direction (referred to the palaeohorizontal) and intensity of NRM of the pilot specimen (○) and one other cylinder (□) from one sample from Nuñorco (a) and La Ojota Formations (b) in response to ac demagnetization. The key to the symbols is as given in Fig. 2.

specimen of each sample from the shales of the Suri Formation. At least one other cylinder from each sample was submitted to the optimum alternating field or temperature in order to improve the definition of the direction of its stable remanence.

A secondary component of remanence of variable intensity was present in all the samples from the Suri Formation; it has the characteristics of a viscous remanent magnetization and it was destroyed by either thermal or ac demagnetization. The optimum demagnetization temperature for the Suri shales was about 400°C, at which temperature more than 35 per cent of NRM intensity was retained for most of the specimens (Fig. 2a); a few samples retained only 5 per cent of their NRM intensity after cleaning to the optimum demagnetization temperature. The optimum demagnetizing ac field for the Suri Tuffs varied between 200 and 225 Oe, at which more than 40 per cent of the NRM was retained (Fig. 2b).

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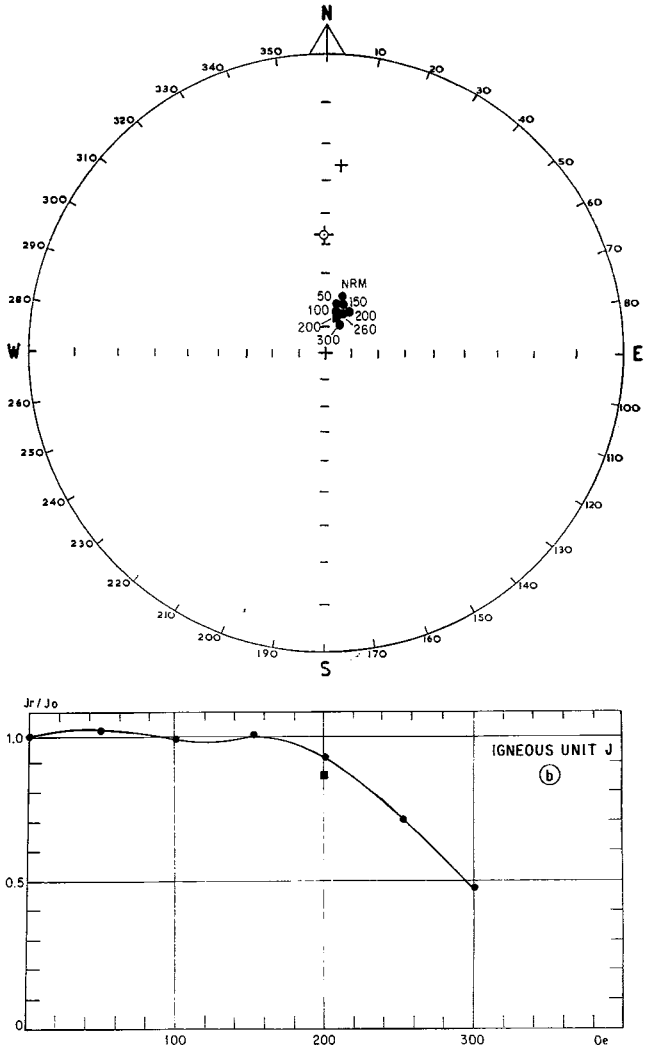


Figure 3 (b)

A secondary component of remanence was also present in most of the samples from the Ñuñorco Formation; it was destroyed by the ac demagnetization (Fig. 3a). The optimum demagnetizing ac field varied between 150 and 200 Oe at which more than 20 per cent of the NRM intensity was retained. Samples from La Ojota Formation have very stable NRM (Fig. 3b).

Fisher's (1953) method was used to compute mean directions of NRM and cleaned remanent magnetization for each stratigraphic level of the Suri Formation (Table 2) and for each igneous unit of the Ñuñorco and La Ojota Formations (Table 3). Unit weights were assigned to the mean direction of natural and cleaned remanences for all the hand samples. For those stratigraphic levels of the Suri Formation and the igneous unit of the Ñuñorco Formation, from which only one sample was collected, the mean direction of remanence was computed, assigning unit weight to the direction of remanence of each of its specimens. Stratigraphic level directions of remanence for the Suri Formation were referred to the palaeohorizontal (Table 2); the fold test could not be applied, however, because all the

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Table 2. Summary of palaeomagnetic data for the Suri Formation (Early Ordovician), north-western Argentina.

Unit and K/Ar age (Myr)	Lithology	Natural remanent magnetization referred to the palaeohorizontal						Cleaned remanent magnetization referred to the palaeohorizontal				South virtual geomagnetic pole	
		N	n	D (°)	I (°)	α_{95} (°)	k	D (°)	I (°)	α_{95} (°)	k	Lat. (°)	Long. (°)E
26	shale	(1)	2	183	24	23	115	176	35	26	94	81 S	87
25	shale	(1)	2	103	18	25	90	96	20	11	525	10 S	15
24	shale	(1)	2	206	−21	12	521	200	−17	7	1160	49 S	143
23	shale	(1)	2	191	−55	61	19	196	−46	25	95	33 S	129
22	shale	(1)	2	35	−15	61	19	76	41	52	24	1 N	355
21	shale	(1)	2	315	32	56	22	224	74	38	45	46 S	262
20	shale	(1)	2	23	−13	25	94	81	44	26	52	4 S	355
19	shale	2	(3)	21	45	>90	1	86	54	66	16	12 S	349
18	shale	(1)	2	311	21	>90	6	39	52	19	160	19 N	326
17	shale	(1)	2	117	52	52	25	110	48	25	54	29 S	2
16	shale	(1)	2	212	22	>90	10	65	45	10	569	7 N	347
15	shale	(1)	2	244	32	4	3131	85	18	11	516	0	11
14	shale	2	(4)	6	−2	57	21	95	50	32	64	18 S	356
13	shale	(1)	2	63	15	41	39	92	39	16	233	12 S	3
12	shale	(1)	2	49	−59	8	961	107	8	5	1928	17 S	26
11	shale	2	(4)	59	−10	13	355	94	29	17	212	10 S	10
10	shale	2	(4)	47	−9	53	24	89	38	10	562	9 S	2
9	shale	2	(3)	330	9	16	230	81	48	39	43	6 S	352
8	shale	2	(4)	30	−42	>90	1	97	44	41	38	17 S	1
7	shale	(1)	2	106	−18	29	78	106	34	17	64	22 S	12
6	shale	(1)	2	104	−24	15	280	106	−5	13	356	13 S	32
5	shale	(1)	2	265	−80	>90	2	121	5	18	63	28 S	35
4	shale	(1)	2	34	−64	32	70	91	3	26	93	1 S	21
3	shale	(1)	2	352	−41	12	403	47	19	36	49	31 S	349
2	shale	(1)	2	330	−40	5	2853	71	25	29	74	10 N	1
1	shale	(1)	2	100	−36	25	100	115	19	55	23	26 S	25
35	tuff	2	(4)	357	−1	22	130	84	27	71	14	1 S	6
34	tuff	2	(3)	27	−11	13	360	50	21	24	107	28 N	351
33	tuff	2	(3)	30	6	88	10	71	22	66	16	11 N	3
32	tuff	2	(2)	12	18	>90	12	86	20	15	265	1 S	11
31	tuff	2	(3)	358	12	>90	1.3	99	34	65	17	16 S	9
30	tuff	3	(3)	53	−20	62	5	78	31	17	51	2 N	2
29	tuff	4	(4)	45	−17	21	20	82	22	8	118	2 N	8
304 ± 10													
28	tuff	3	(3)	52	−16	>90	3	91	35	37	12	10 S	5
27	tuff	3	(6)	96	26	30	17	99	34	16	59	16 S	9

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Table 3. Summary of palaeomagnetic data for igneous rocks from Ñuñorco (Late Silurian–Early Devonian) and La Ojota (Middle Silurian) Formations, north-western Argentina. Data for units D and E referred to the present horizontal. References for K–Ar ages are in Table 1 and Linares (1977).

Formation	Unit and K/Ar age (Myr)	Lithology	Natural remanent magnetization referred to the palaeohorizontal						Cleaned remanent magnetization referred to the palaeohorizontal				South virtual geomagnetic pole	
			N	n	D (°)	I (°)	α_{95} (°)	k	D (°)	I (°)	α_{95} (°)	k	Lat. (°)	Long. (°)E
La Ojota	J 297 ± 10	dyke	5	(10)	26	64	7	124	32	68	7	127	6 S	312
	F	igneous rock	4	(8)	197	60	>90	1.2	198	77	28	12	51 S	280
	G_1	igneous rock	(1)	2	26	43	66	31	25	40	38	44	34 N	320
	G_2	(20 m lower than F)	(1)	2	101	49	15	283	94	57	25	98	19 S	349
	H	igneous rock (20 m lower than G)	2	(4)	18	−2	9	777	335	56	22	129	21 N	270
	I	igneous rock (10 m lower than H)	2	(4)	101	6	19	167	101	−3	44	34	9 S	28
Ñuñorco	E	contact baked by dyke D	4	(8)	46	53	7	36	97	67	19	24	25 S	337
	D 360 ± 10	rhyolitic dyke	5	(10)	79	60	75	2	122	74	20	16	39 S	325
	C	andesite	2	(4)	313	81	>90	5	114	69	14	332	36 S	336
	B 416 ± 25	rhyodacite	4	(8)	342	10	10	83	17	38	15	36	38 N	312
	A 310 ± 10	red porphyry	3	(6)	300	20	>90	1.5	334	−28	35	13	63 N	224

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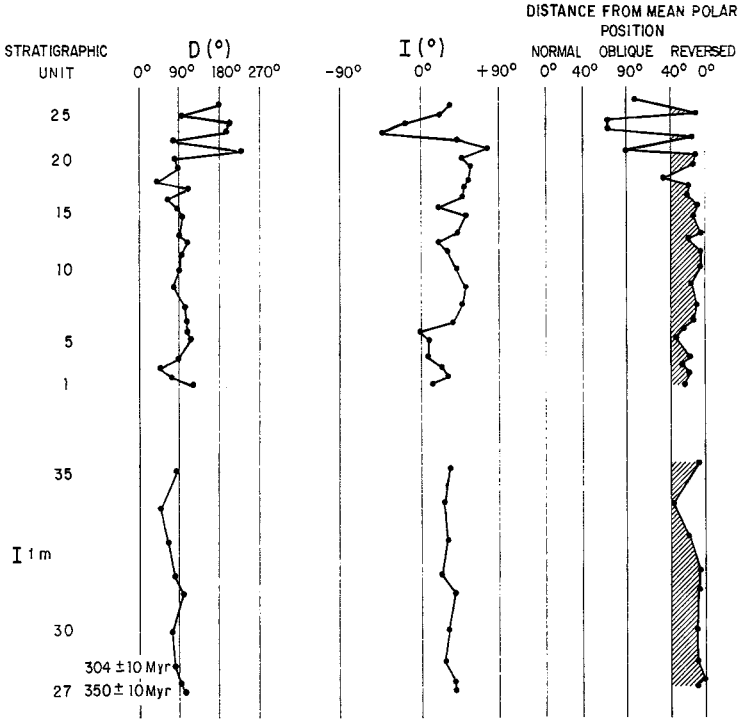


Figure 4. Suri Formation. Mean declination (D) and inclination (I) of stable remanence (Table 2) and the angle between the VGP for each stratigraphic unit and the palaeomagnetic pole through the sequence of Suri rocks.

samples were collected from a sequence where the dip of the strata is uniform (dip, 24°; strike 358° for the tuffs and dip, 18°; strike 332° for the shales).

Virtual geomagnetic poles (VGP) for each stratigraphic level of the Suri Formation were calculated (Table 2). A mean pole was calculated from these VGP's; VGP's situated more than 40° from the mean pole were rejected and a new mean pole was calculated. In this way a population of VGP's was obtained, all within 40° of the mean pole; this population of VGP's yields a palaeomagnetic pole for the Suri Formation at 5.9° E, 8.5° S ($N = 30$, $k = 20.7$, $\alpha_{95} = 5.9°$, $R = 28.6$, $SAO1_1$). The rejected VGP's were classified as oblique.

The magnetic stratigraphy for the Suri Formation, consisting of logs of stratigraphic level – mean values of cleaned declination (D) and inclination (I), is plotted alongside the log of distances of VGP's from the mean polar position (Fig. 4). It should be noted that all except five (obliques) of the stratigraphic units yield reversed mean cleaned remanence directions. The character of variations in D and I along the tuffs does not differ from that along the shales: D and I undergo relatively small oscillations around 100° and 35°, respectively. This relatively quiet behaviour holds true for the Suri Formation, except at the levels of units 21–26, where reversals of both D and I are recorded.

All but two of the igneous units of Ñuñorco and La Ojota Formations yield cleaned remanence directions with positive inclination (Table 3). In particular, the red porphyric unit of the Ñuñorco Formation, exposed close to Puesto Chaschuil, yielded northerly declination and negative inclination; the K–Ar age for this igneous unit is 310 ± 10 Myr (Linares 1977). VGP's for each igneous unit of the Ñuñorco and La Ojota Formations were calculated (Table 3). The positions of VGP's for the Ñuñorco Formation are scattered; however, the positions of VGP's C, D, E and G_2 and B and G_1 seem to define two groups.

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These groups yield mean poles at 337° E, 30° S ($N=4$, $k=41$, $\alpha_{95}=14°$, $R=3.93$) and 316° E, 36° N ($N=2$, $k=227$, $\alpha_{95}=17°$, $R=1.99$).

5 Interpretation of results

If at least one VGP lies within 40° of the normal palaeomagnetic pole we suggest that the succession of oblique VGP's indicates a short polarity event within a reversed sequence of rocks (Valencio *et al.* 1977). The other successions of oblique VGP's are interpreted as records of geomagnetic excursions; accordingly we interpret the reversals of D and I recorded at the top of the Suri rocks sequence (Fig. 4) as geomagnetic excursions.

Khramov *et al.* (1978) have reported reversed and normal remanences in USSR Early and Late Llanvirnian rocks, respectively (North Baikal Superzone). The reversed and oblique

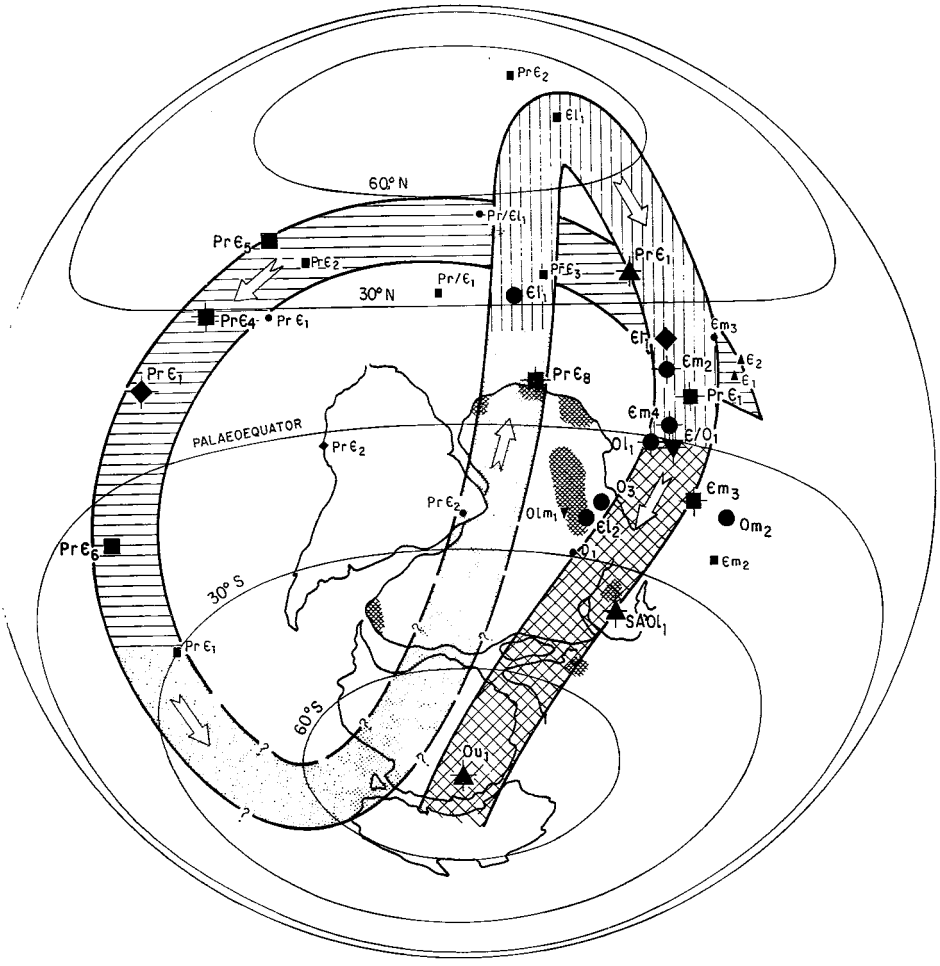


Figure 5. Pole position for the Suri Formation (SAO1$_1$, Llanvirnian age, older than 416 ± 25 Myr) plotted on the Vilas & Valencio (1978b) reconstruction. Pole positions for the Gondwana continents are those quoted in Vilas & Valencio (1978a): △, South America; □, Africa; ○, Australia; ◇, India; ▽, Antarctica. The key to these symbols is as given in the last quoted paper (pole prefix omitted from the figure). Lambert's equal-area projection. The age of the tentative polar path is as follows: ▤ Precambrian, from 1000 to 700 Myr; ▨ Late Precambrian; ▥ Cambrian; ▩ Ordovician. The Late Ordovician glaciations are also shown.

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nearer normal cleaned remanences isolated in the Suri rocks are consistent with those reports, suggesting that the tuffs and the lowest shales of the Suri Formation are Early Llanvirnian.

Following the classification system for palaeomagnetic data suggested by Vilas & Valencio (1978a), we have allocated six points (key pole) to the Suri palaeomagnetic pole position ($SAO1_1$). Fig. 5 shows the position of $SAO1_1$ plotted on the Vilas & Valencio (1978b) reconstruction of Gondwana. The figure also shows the palaeomagnetic pole positions for the Gondwanic continents for the period 1000–400 Myr ago and the joint apparent polar wandering path suggested for the Gondwana Supercontinent for that period by the first quoted authors. An examination of Fig. 5 shows that the position of $SAO1_1$ is in relatively good agreement with the positions of palaeomagnetic poles for the Tumblagooda sandstone (AuO_3; Ordovician; 5 points: main pole) and Stairway sandstone ($AuOm_2$; Middle Ordovician; 5 points: main pole) of Australia and for the Sør Rondane intrusives ($AnO1\text{-}m_1$, Early–Middle Ordovician, 485 ± 25 Myr, 4 points: subsidiary pole) of Antarctica (references quoted in Vilas & Valencio 1978a). This is consistent with the interpretation that the supercontinent of Gondwana existed as far back as 1000 Myr ago (McElhinny & Embleton 1976; Vilas & Valencio 1978a). Nevertheless, the distances defined between the positions of those poles, (as well as the errors in polar positions and the inaccuracy of the age assigned to the rocks) do not exclude geodynamic models involving the formation and destruction of minor oceans.

Fig. 5 shows that the positions of the Ordovician palaeomagnetic poles for the Gondwana Supercontinent are consistent with the Late Ordovician glacial deposits in Arabia, India and Africa.

The mean position of the VGP's for the igneous units C, D, E and G_2 of the Late Silurian–Early Devonian age Ñuñorco Formation (item 4) is consistent with the Late Ordovician to Late Carboniferous section of the joint apparent polar wandering path for Gondwana suggested by Vilas & Valencio (1978a, see Fig. 4 of that paper). It must be noted though that the palaeomagnetic data for only four of the 10 igneous units of the Ñuñorco Formation were used to calculate this mean position. We do not offer a sound interpretation to explain the scattering of the Ñuñorco's VGP's. The stability tests and the predominant reversed polarity suggest, however, that the stable remanence of Ñuñorco rocks might record the geomagnetic field in Late Silurian–Early Devonian times. It should be appreciated though that tectonic complexities associated with formations integrated only by igneous rocks are frequently high and not easily discovered during the field work. These complexities could be the cause of the scattering of the palaeomagnetic data for the Ñuñorco rocks. Accordingly these data are not used for the interpretation of the geodynamic history of either South America or the Gondwana Supercontinent in early Palaeozoic.

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